



King Saud University
Saudi Journal of Ophthalmology

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ORIGINAL ARTICLE

Uveal melanoma in the Saudi Arabian population: Two decades of management at the King Khaled Eye Specialist Hospital

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Received 31 May 2009; accepted 31 May 2009
Available online 5 August 2009

KEYWORDS

Uveal melanoma;
Choroidal melanoma;
Enucleation;
Plaque radiotherapy;
King Khaled Eye Specialist
Hospital;
Saudi Arabia

Abstract *Objective:* To present the experience of King Khaled Eye Specialist Hospital (KKESH) with uveal melanoma over the last two decades in a fashion similar to the result of the Collaborative Ocular Melanoma Study (COMS).

Design: Retrospective, non-comparative, interventional, case series.

Participants: All patients were diagnosed with uveal melanoma at the King Khaled Eye Specialist Hospital (KKESH), Riyadh, Saudi Arabia from June 1983 to July 2005 and met the inclusion criteria of the COMS.

Methods: A medical record review of clinical history, imaging studies, surgical procedures and treatment outcome was performed.

Results: Forty patients (24 males and 16 females) with uveal melanoma (average age 50 years; range 24–77 years) were included in the study; 28 (70%) were of Saudi Arabian descent and the remaining 12 (30%) patients were from neighboring Arab countries. Decreased vision was the main presenting complaint of 29 (72.5%) patients; the duration of this symptom was 3 months or more in 27 (67.5%) patients. The apical height of the tumor was 10 mm or more in nine (22.5%) of the affected eyes and the largest basal dimension was more than 16 mm in nine (22.5%) of the affected eyes. The posterior border of the tumor was 1–2 mm from the optic disc in three (7.5%) affected eyes. Primary enucleation was performed for 33 (82.5%) eyes, episcleral radiation plaque therapy for six (15%) of the eyes and endo resection of the uveal melanoma in one (2.5%) eye. Adjunct external beam radiation

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therapy was performed in two (5%) orbits for extrascleral extension. The histopathological diagnosis was available for 34 (84%) eyes in which surgery had been performed (33 patients underwent primary enucleation and one patient underwent endo resection of the uveal melanoma); 24 (70.6%) eyes had spindle cell and the remaining 10 (29.4%) had epithelioid or mixed cell types. Evidence of extraocular tumor extension was found in three eyes. The average follow-up was 33.7 months with a median of 19 months (range 0.5 months to 10 years). Two (5%) patients developed metastasis after 2 years and 5 years from the initial treatment of large and medium-sized uveal melanomas, respectively.

Conclusion: Individuals of Saudi Arabian ancestry appear to have a low incidence of uveal melanoma. Further studies are required to estimate the true incidence of uveal melanoma in the larger Arab population. Early detection is essential for improving the patient outcomes. Regular communication between the tertiary care eye centers and the local ophthalmic care providers is required to enhance the understanding about uveal melanoma behavior in Arab population.

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Conflict of interest

The author has no conflicts of interest or proprietary interest in any of the instruments or topics presented in this article.

1. Introduction

Uveal melanoma is the most common primary intraocular malignancy in adults. Although, uveal melanoma is not as aggressive as cutaneous melanoma, it is associated with a substantial risk for metastasis (Zakka et al., 1980). Enucleation has been the standard treatment for eyes with uveal melanoma in the past. However, despite treatment, the 5-year prognosis remains poor, with mortality rates ranging from 2% to 50%, depending on the tumor size (Diener-West et al., 1992). Over the past three decades radiotherapy and other treatment approaches with the goal of improving survival and preserving the eye have become increasingly popular. Despite advances made with local methods of treatment of uveal melanoma, an improvement in the rate of survival of patients has not been achieved (Singh and Topham, 2003).

The Collaborative Ocular Melanoma Study (COMS) has provided a unique opportunity to assess a large series of patients with choroidal melanoma. Participating patients are enrolled, evaluated prospectively, treated, and followed in a standardized fashion (Collaborative Ocular Melanoma Study Group, 1993). The enrollment of patients in the COMS started in November of 1986. The enrollment ended in 1998 for medium-sized tumors, in 1994 for large tumors and in 1989 for small tumors (Singh and Kivelä, 2005; Melia et al., 2001). The COMS tried to exclude patients with other co-morbidities that may cause earlier death to provide the longest possible follow-up.

Review of the medical literature showed that uveal melanomas have been extensively studied in the western hemisphere. However, there are scarce reports discussing uveal melanomas from other parts of the world. The purpose of this study was to present the experience with uveal melanoma management, over the last two decades, at the King Khaled Eye Specialist Hospital (KKESH) in Riyadh, Saudi Arabia, with a predominant Arab population. This experi-

ence will be presented using the structure developed by the COMS.

2. Patients and methods

After obtaining approval from the institutional review board, medical records of all patients with the diagnosis of uveal melanoma seen at the KKESH from June 1983 to July 2005 were retrospectively reviewed. In this study, only the patients who met the eligibility criteria for the COMS were included (Box 1) (Singh and Kivelä, 2005). This is to present our experience in a fashion that can be compared to the reported results of the COMS. The parameters studied included gender, age at the time of presentation, initial signs and symptoms, significant associated ocular and systemic disease, visual acuity (VA), and intraocular pressure (IOP). The tumor clinical characteristics studied included tumor location, tumor meridian, distance of posterior edge of the tumor from the optic nerve and fovea, maximal basal tumor diameter, thickness, and shape, as well as the presence of subretinal fluid, laterality, location, and the size of the tumor determined by clinical examination.

In our review, the tumor was classified as uveal melanoma using the COMS classification based on the tumor dimensions. The COMS divided uveal melanomas based on size into small, medium and large tumors (Singh and Kivelä, 2005; Collaborative Ocular Melanoma Study Group, 1995). A small melanoma was 5–16 mm at the largest basal diameter (LBD) and 1.0–2.5 mm in apical height. Tumors that were smaller than 1.0 mm in apical height and less than 5 mm in LBD were regarded as probable choroidal nevi. A medium-sized melanoma was 16 mm or less at the LBD and had an apical height between 2.5 and 10.0 mm, unless the tumor was located within 2.0 mm of the optic disc, which is not an easy site for placement of the brachytherapy. Uveal melanomas more than 16.0 mm at the LBD, more than 10.0 mm in height or located within 2.0 mm of the optic disc were defined as large tumors. Information from the medical records was retrieved regarding treatment, complications of the treatment, and histopathological findings in cases of enucleation. The histopathology was further reviewed regarding extraocular extension of the tumor such as extrascleral and/or optic nerve involvement at the time

of enucleation. The uveal melanomas were divided into spindle cell, mixed cell or epithelioid cell types. Follow-up of these patients where available was documented including evidence of metastasis at the time of the last follow-up. Since the KKESH is a tertiary care facility, after definitive treatment some of the patients were followed up by their referring facilities. Hence, long-term follow-up was not possible for all patients.

Box 1 Eligibility criteria for the COMS.

Primary choroidal melanoma in one eye
 Less than 50% involvement of the ciliary body
 Age 21 years or older
 Ability to give informed consent
 Ability to return for treatment and scheduled follow-up
 No other primary cancer or history of cancer other than non-invasive non-melanotic skin cancer or carcinoma in situ of the uterine cervix
 No coexisting disease threatening survival for 5 years or longer
 No metastatic melanoma
 No contraindication for surgery or radiation (in medium-sized and large tumor randomized trials)
 No previous fine-needle aspiration biopsy of choroidal melanoma
 No previous treatment of choroidal or ciliary body melanoma or any condition secondary to the tumor
 No extrascleral extension of 2.0 mm or more.
 No diffuse, ring, or multifocal tumor. No iris or angle involvement by tumor
 No use of immunosuppressive therapy that could not be discontinued

3. Results

There were 60 eyes in 60 patients diagnosed with uveal melanoma from June 1983 to July 2005 at the KKESH. Forty patients met the inclusion criteria. Among the 20 patients who did meet the inclusion criteria the following characteristics were noted. Eight patients had uveal melanoma mainly involving the ciliary body, one patient had melanoma of the iris, six patients had extrascleral extension of more than 2 mm at presentation, three patients had diffuse tumor, one patient had biopsy of the tumor before presentation to our institution, and one patient was younger than 21 years of age.

The average age at presentation was 50 (range 24–77 years). There were 24 males and 16 females with a male to female ratio of 1.5:1. Of the patients, 28 (70%) were Saudi Arabian and the remainder were from neighboring Arab countries. The right eye was affected in 18 (45%) patients and the left eye in 22 (55%) patients. There was no known significant associated systemic disease.

The presenting symptoms included decreased vision alone in 29 (72.5%) patients, decreased vision associated with ocular pain in five (12.5%) patients and floaters in five (12.5%) patients. One (2.5%) of the tumors was discovered incidentally. Thirteen patients (32.5%) presented within 2 months of the start of symptoms, 17 (42.5%) patients presented 2–6 months from the start of symptoms, nine (22.5%) patients presented 6–12 months from the onset of symptoms, and one (2.5%) pa-

tient presented more than 12 months from the start of symptoms.

At presentation, the VA in the involved eye was 20/20 to 20/60 in 10 (25%) eyes, 20/70 to 20/200 in 7 (17.5%) eyes, CF in 6 (15%) eyes, HM to LP in 11 (27.5%) eyes and NLP in 2 (5%) eyes. The average intraocular pressure (IOP) was 19 mmHg (range 8–64 mmHg), and nine (22.5%) eyes had an IOP of 24 mmHg or higher. The range of LBD of the eyes with melanoma was between 5 mm and 26 mm. The LBD was more than 16 mm in nine (22.5%) eyes. The range of the apical height was between 3.2 mm and 16 mm. The apical height was more than 10 mm in nine (22.5%) eyes (Table 1). The posterior border of the tumor was 1–2 mm from the optic disc in three (7.5%) eyes. According to the COMS classification, 20 (50%) of our patients had large tumors, 20 (50%) patients had medium tumors, and none of our patients had small tumors. The tumor involved the macula in four (10%) eyes. Subretinal fluid was present in 35 (87.5%) eyes. A neovascular glaucoma (NVG) was found in five (12.5%) eyes with uveal melanoma, all of these eyes had an IOP of 25 mmHg or higher.

In all eyes, the diagnosis of uveal melanoma was made on clinical and U/S findings, and it was confirmed histopathologically in those eyes that were enucleated. Enucleation was performed in 33 (82.5%) eyes. Six (15%) eyes were treated with episcleral I¹²⁵ radiation plaque therapy, and in one eye endoresection was performed with simultaneous episcleral I¹²⁵ radioactive plaque therapy. Two (5%) patients received additional external beam radiation treatment due to the presence of extraocular tumor extension found on histopathological examination of the enucleated eyes and one patient refused the radiation treatment.

A definite diagnosis of uveal melanoma was confirmed by histopathology in 34 eyes. On histopathological examination, 24 (70.6%) tumors were found to have spindle cell type uveal melanoma and the remaining 10 (29.4%) tumors had mixed (<50% of cells epithelioid in type) or epithelioid cell types of (>50% of cells epithelioid in type) uveal melanoma (Table 2). Extrascleral extension was found in three eyes, while no optic nerve invasion was identified in any of the patients.

Metastatic uveal melanoma was detected in two (5%) patients, after 2 years and 5 years from the initial treatment of large and medium-sized uveal melanomas, respectively. The site of metastasis was the liver in both patients. Because KKESH is a tertiary care facility, the average follow-up was 33.7 months with a median of 19 months (range 0.5 months–10 years). During the same period, that is June 1983–July 2005, there were 567 new cases of retinoblastoma diagnosed and treated at the KKESH.

4. Discussion

In the United States, about six cases per million are diagnosed annually, accounting for 1600–2000 new cases every year (Dienner-West et al., 1992). Although we do not have a well-developed registry for uveal melanoma in Saudi Arabia, 60 cases of uveal melanoma were documented between 1983 and 2005; these were compared with 567 cases of retinoblastoma at KKESH during the same period. The KKESH is a tertiary referral center for eye diseases in the region, and the registered cases provide the impression that uveal melanoma occurs with a low frequency in this region. A similar finding was reported

Table 1 Baseline characteristics of small, medium-sized, and large melanomas for the current study and the COMS.^a

Clinical feature	Small tumor (%)		Medium-sized tumor (%)		Large tumor (%)	
	Current study	The COMS	Current study	The COMS	Current study	The COMS
<i>Sex</i>						
Male:female	—	50:50	65:35	49:51	55:45	57:43
<i>Race</i>						
White, not hispanic	—	99	—	98	—	97
Saudi	—	—	60	—	80	—
<i>Age</i>						
< 40 years	—	9	55	9	10	9
40–49 years	—	11	5	14	30	13
50–59 years	—	19	25	20	5	21
60–69 years	—	33	20	31	45	29
70–79 years	—	22	0	21	10	23
≥ 80 years	—	6	0	4	0	5
<i>Anterior tumor border</i>						
Iris	—	0	0	0	0	0
Ciliary body	—	0	10	0	20	6
Pars plicata	—	2	0	4	0	23
Pars plana	—	3	0	8	0	21
Equator to ora serrata	—	22	70	34	55	36
Posterior to equator	—	73	20	54	25	12
<i>Posterior tumor border</i>						
Ciliary body	—	0	0	0	0	0
Equator to ora serrata	—	6	20	2	5	3
Posterior to equator	—	94	80	98	95	97
<i>V/A of tumor eye</i>						
≥ 20/20	—	44	10	33	0	12
20/25–20/40	—	38	15	37	5	14
20/50–20/160	—	15	35	19	5	40
≤ 20/200	—	3	40	10	90	33

^a The baseline characteristics of small, medium-sized, and large melanomas for the COMS taken from Singh and Kivelä (2005).

Table 2 Histopathological features of medium-sized and large melanomas in current study and the COMS.^a

Histopathological features	Medium-sized tumor		Large tumor	
	Current study (%)	The COMS (%)	Current study (%)	The COMS (%)
<i>Cell type</i>				
Mixed cell type	0	84.7	30	86.7
Spindle cell type	93	12.3	55	7.2
Epithelioid cell type	7	3	15	6.1
Neovascularization of the iris	10	2	15	8
Optic nerve invasion	0	0	0	7
Extrascleral extension	5	3	10	8

^a Histopathological features of medium-sized and large melanomas in the COMS were taken from Collaborative Ocular Melanoma Study Group (1998a).

from the Shanghai Eye, Ear, Nose and Throat Hospital in China between 1955 and 1979, where there were 103 cases of uveal melanoma compared to 631 cases of retinoblastoma (Kuo et al., 1982). By contrast, 688 cases of ocular melanoma compared to 110 cases of retinoblastoma were diagnosed among New York State residents between 1975 and 1986 (Mahoney et al., 1990). Despite the fact that there are limited population-based statistics on the incidence of uveal melanoma according to race and ethnicity, there appears to be racial variation for the occurrence of uveal melanoma. Although reports

on the incidence of uveal melanoma in Asian population are very scarce, the population-based incidence of uveal melanoma, in Asian patients living in Western countries, has not been found to be significantly different from that reported in other non-Caucasian patients studies. From the National Cancer Institute's data on 1352 patients with uveal melanomas, diagnosed in the United States from 1992 to 2000, the calculated age- and race-adjusted annual incidence per million was 0.31 in African-Americans, 0.38 in Asians, 1.67 in Hispanics, and 6.02 in the non-Hispanic Caucasian population (Hu

et al., 2005). From the 7206 patients with uveal melanoma, seen by the 43 COMS participating Clinical Centers during the enrollment period, 2524 eligible patients agreed to be enrolled in the randomized study (Singh and Kivelä, 2005). Among the enrolled patients in the COMS randomized trials, less than 2% were non-Caucasians (Collaborative Ocular Melanoma Study Group, 2001a). Racial differences in the incidence and distribution of benign and malignant melanotic tumors of the skin are well documented; the occurrence of these tumors in non-white populations appears to account for less than 0.2 of the incidence in whites (Krementz et al., 1976).

Uveal melanoma is a disease of adults with a median age at initial diagnosis of around 53 years (McLean, 1996). In a study of 6358 cases of uveal melanoma, 1.6% of the patients were found to be less than 20 years of age at the time of diagnosis (Barr et al., 1981). Patients in our study presented at an average age of 50 years. The average age, at presentation, was around 10 years more in patients who participated in the COMS than in our patients (Collaborative Ocular Melanoma Study Group, 2001b). There was a predominance of males in our retrospective study, consistent with most reported studies (McLean, 1996; Barr et al., 1981; Scotto et al., 1976; Phillpotts et al., 1995). However, a male predominance was not found in the COMS randomized prospective study (Collaborative Ocular Melanoma Study Group, 2001a).

The size of the tumor is a well-known prognostic factor in patients with uveal melanoma (Mooy and De Jong, 1996; McLean et al., 1980). Therefore, accurate assessment of tumor size is essential for treatment selection such as observation, resection, radiation or enucleation. The accuracy of diagnosing uveal melanoma, based on clinical and imaging techniques involving U/S, CT-scan, MRI, and angiography, has improved significantly in recent years. This fact has been supported by the findings of the COMS, where the accuracy of correctly diagnosing a medium-sized choroidal melanoma was found to be 99.5%. For large choroidal melanomas, the accuracy was reported to be over 99.7% with histopathological confirmation of the diagnosis (Collaborative Ocular Melanoma Study Group, 1990). All 34 patients in our study, who had histopathological evidence of uveal melanoma, were diagnosed as such before confirmation by histopathology.

In our review, a significant number of patients had a visual acuity of 20/200 or worse and evidence of NVG compared to the patients in the COMS (Singh and Kivelä, 2005; Collaborative Ocular Melanoma Study Group, 1998a). This finding might be explained by the delay in presentation of more than 2 months, after the onset of ocular complaints, in 67.5% of patients. The majority of the COMS patients in the medium and large melanoma groups were enrolled within a month from the diagnosis (Singh and Kivelä, 2005). However, the actual time between the initial ocular complaints and the diagnosis was not reported by the COMS.

According to the modified Callender's classification, spindle cell tumors carry the best prognosis and epithelioid cell tumors the worst (McLean et al., 1983). Among the uveal melanoma cases reported, the mixed cell type tumors are the most common, followed by the spindle cell type, while the epithelioid type tumors are least common (McLean, 1996; Collaborative Ocular Melanoma Study Group, 1998a). Among our patients, the spindle cell type was found in 70.6% of patients followed by the mixed cell type in 17.6% of patients, and epithelioid type in 11.8% of patients. In the COMS, in contrast to our

study, most tumors (86%) were the mixed cell type (McLean et al., 1983).

Many studies have found that tumor size is the strongest indicator of metastasis and survival (Mooy and De Jong, 1996). The estimated melanoma-related mortality was 1% at 5 years and was 4% at 8 years for patients with small melanomas in the COMS (Collaborative Ocular Melanoma Study Group, 1997a). The 5-year melanoma-related mortality, based on histopathologically confirmed metastasis, increased to 11% for enucleation alone and was 9% for brachytherapy among patients with medium-sized tumors in the COMS (Collaborative Ocular Melanoma Study Group, 2001a). The 5-year mortality based on histopathologically confirmed metastasis increased even more with large tumors and was 28% for cases in whom enucleation alone was performed and 26% for cases where pre-enucleation radiation was given (Collaborative Ocular Melanoma Study Group, 1998b). The delay in presentation to our institution likely played a major role in finding a significant number of patients with large uveal melanomas. A comparative study well controlled for age and the gender of patients, site of the primary lesion, stage of disease at manifestation, and Clark's level of invasion of the primary cutaneous melanoma, showed that the prognosis was significantly worse in black patients than in Caucasians (Reintgen et al., 1982). This is in contrast to the studies on uveal melanoma where the prognosis is no worse in black patients than it was in Caucasian patients (Margo and McLean, 1984). Due to the very short follow-up period available (19 months or less for 50% of patients), it was difficult to determine the survival outcome among our patient population.

Metastasis is common following a primary diagnosis of uveal melanoma; frequently, the site of metastasis is the liver. In patients who developed metastases, liver involvement ranged from 56% to 100% (Collaborative Ocular Melanoma Study Group, 2004). Of the patients enrolled in the COMS randomized trials, 739 patients were diagnosed with at least one site of melanoma metastasis during the follow-up after primary treatment of a choroidal melanoma (Collaborative Ocular Melanoma Study Group, 2004). Overall Kaplan-Meier estimates of the 2-, 5- and 10-year metastasis rates were 10%, 25% and 34%, respectively (Collaborative Ocular Melanoma Study Group, 2004). In the COMS, the liver was the predominant site of metastasis, which was reported in 89% of patients (Collaborative Ocular Melanoma Study Group, 2004). Annual screening for metastasis with liver function tests and chest radiograms was performed in the COMS. Since the use of liver function tests followed by other diagnostic tests had a low sensitivity, liver imaging is advocated by some physicians to identify metastatic disease associated with choroidal melanoma earlier (Collaborative Ocular Melanoma Study Group, 2004, 2001c; Eskelin et al., 1999; Eskelin and Kivela, 2002). The data from the COMS showed that the median time from diagnosis of metastasis to death was less than 6 months. The death rates were 80% and 92% at 1 and 2 years following a report of melanoma metastasis (Collaborative Ocular Melanoma Study Group, 2004). There was no statistically significant difference in the tumor dimensions of subgroups with respect to time from diagnosis of metastasis to death (Collaborative Ocular Melanoma Study Group, 2005). Patients who developed metastasis in our study were referred to general hospital for further management and no follow-up data were available.

Given the observed correlation between the increased metastasis rate with larger choroidal melanomas at the time of enrollment and primary treatment, the COMS findings suggested the need for earlier diagnosis and treatment of patients with primary choroidal melanomas. Small uveal melanomas may be safely observed. The presence of orange pigment, larger tumor dimensions and the lack of drusen and adjacent retinal pigment epithelial changes mandate closer observation; these characteristics were identified as risk factors predictive of growth of small uveal melanomas (Collaborative Ocular Melanoma Study Group, 1997b). The COMS data suggest that the mortality rates among patients treated with eye-conserving I¹²⁵ brachytherapy were similar to the mortality rates among patients treated with enucleation for up to 12 years of follow-up. This was based on deaths from all cause-mortality as well as tumor-specific mortality (Collaborative Ocular Melanoma Study Group, 2006). The COMS findings support that patients with a choroidal melanoma, of the appropriate size, and who meet other COMS eligibility criteria, are good candidates for I¹²⁵ brachytherapy without compromising the survival outcome. However, these patients must be advised about the possibility of visual loss and subsequent enucleation due to tumor growth or other complications. The COMS findings showed that pre-enucleation irradiation did not improve the survival of patients with a large uveal melanoma within the first 8 years of follow-up (Collaborative Ocular Melanoma Study Group, 1998b). However, since irradiation reduced mitosis in the tumor of the pre-enucleated eyes, long-term follow-up will be necessary to exclude a late survival difference (Collaborative Ocular Melanoma Study Group, 1998a; Manschot and van Strik, 1992).

The COMS data suggest that earlier diagnosis and treatment of choroidal melanoma and better methods for the detection and treatment of melanoma metastasis are necessary to improve the long-term prognosis for these patients. Better understanding of the body's immune response to a uveal melanoma along with the cytogenetics and genomics of uveal melanoma might lead to more effective and less invasive therapeutic intervention.

Although Arab populations may have a low incidence of uveal melanomas compared to Caucasian populations, unfortunately late presentation to the eye care specialist in a majority of cases leads to significant visual morbidity and mortality. Therefore, with increased awareness, improvement in early detection and prompt referral by the general ophthalmologist might help preserve the eyes and survival of affected patients. Since this study was a retrospective chart review the follow-up after treatment was limited. Development of our new national tumor registry should help identify new cases of uveal melanoma and improve our incidence and outcome statistics in Saudi Arabia. Continuous communication before and after receiving treatment at the tertiary eye centers is essential for collecting the information needed on uveal melanoma to determine more accurate information about this disease in the Arab population.

Acknowledgements

The author would like to thank Dr. Emad Abboud, Dr. Imtiaz Chaudhry, Dr. Essam Al-Harthi and the Research Depart-

ment, King Khaled Eye Specialist Hospital for their great help in this project.

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